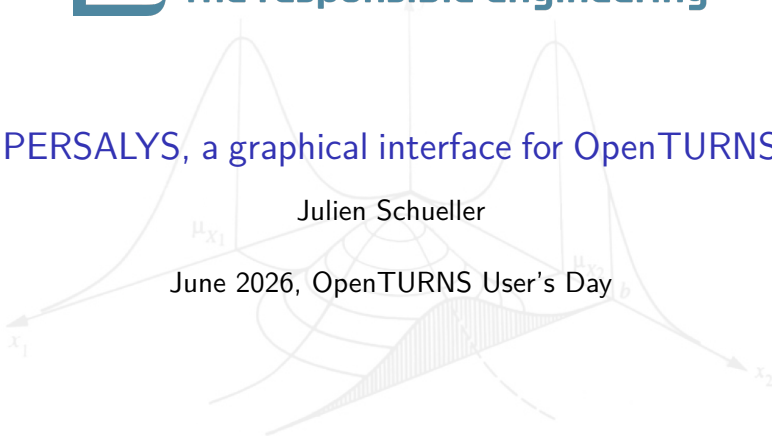


PERSALYS, a graphical interface for OpenTURNS

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June 2026, OpenTURNS User's Day



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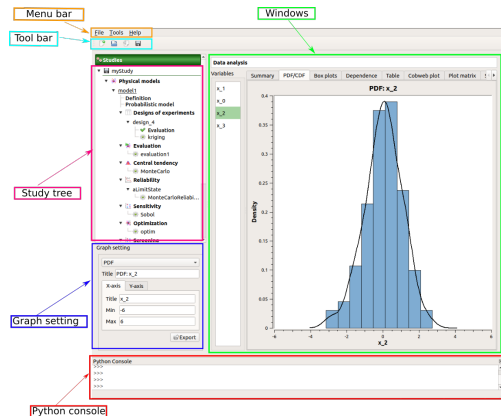
v19 / v19.1

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Summary & conclusion

What is Persalys?

- ▶ Graphical interface for OpenTURNS, developed jointly by **EDF R&D** and **Phimeca** since 2015
- ▶ Open source (LGPL)
- ▶ Stand-alone or embedded in the Salome platform (HPC)
- ▶ Key capabilities:
 - ▶ Uncertainty quantification (distribution fitting, copulas)
 - ▶ Sensitivity analysis (Sobol, Morris, HSIC)
 - ▶ Metamodeling (kriging, polynomial chaos, regression)
 - ▶ Reliability (FORM/SORM, Monte Carlo)
 - ▶ Optimization, calibration
 - ▶ External code coupling (Python, FMI, YACS, coupling model)

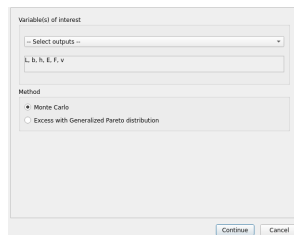


Releases overview (2025–2026)

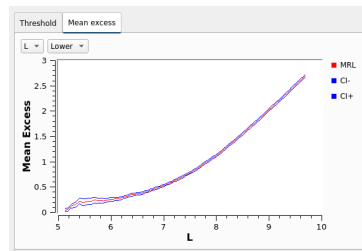
- ▶ **18.0** (Apr. 2025) – Quantile analysis, data field model
- ▶ **18.1** (Jun. 2025) – Metamodel error measures, CSV improvements
- ▶ **19.0** (Oct. 2025) – Data sets sensitivity, DOE export/import, Student copula, histogram distribution
- ▶ **19.1** (Dec. 2025) – SSH remote coupling, regression improvements, ergonomics
- ▶ **20.0** (Mar. 2026) – HSIC sensitivity, workflow reorganization, Python package installer
- ▶ **20.1** (Jun. 2026) – Miscellaneous improvements, bug fixes

v18: Quantile analysis

- ▶ New analysis available on **data models**
- ▶ Estimate quantiles for given probability levels (low tail, high tail, bilateral)
- ▶ Two methods:
 - ▶ **Monte Carlo**: empirical quantile + confidence interval
 - ▶ **Excess method**: fit a Generalised Pareto Distribution to the tail; mean excess plot to choose the threshold



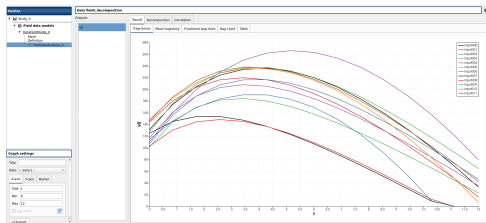
Quantile analysis				
Output		Quantiles	CDF	
		Target probability	Type	Quantile Confidence interval @95%
L	0.01	$P(L < q) = 0.01$	5.54111	[5.46588; 5.61634]
		$P(L > q) = 0.99$	8.88254	[8.73497; 9.03011]
F	0.05	$P(L < q) = 0.05$	5.9161	[5.8693; 5.9631]
		$P(L > q) = 0.95$	8.17745	[8.10737; 8.24753]



v18: Data field model – v18.1: Metamodel improvements

Data field model (v18)

- ▶ Load a file of 1-D field trajectories as a data model
- ▶ Mesh defined in Persalys or loaded from file
- ▶ Central tendency analysis (Karhunen-Loève decomposition) available on imported field data

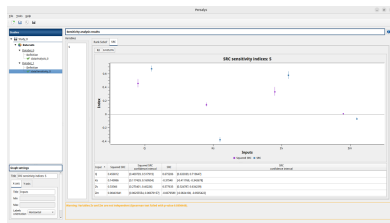
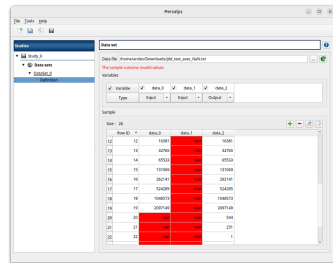


Metamodel improvements (v18.1)

- ▶ Consistent error measures: MSE and R^2 unified across kriging, polynomial chaos and linear regression
- ▶ New CSV parser (OpenTURNS 1.25): supports missing values and unicode characters
- ▶ FCE metamodel from data model now exports the probabilistic model as well

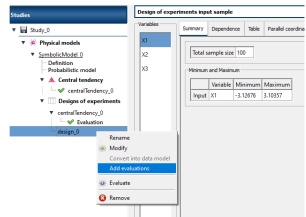
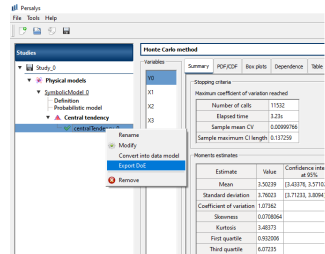
v19: Data sets – sensitivity analysis & missing values

- ▶ “Data models” renamed to **data sets**
- ▶ Data analyses can now be performed on samples containing **missing / NaN values**
 - ▶ Marginal analyses use available data per variable
 - ▶ Multivariate analyses use the valid sub-sample
- ▶ **Sensitivity analysis** available directly on data sets: SRC indices and Sobol' indices via RankSobolSensitivityAlgorithm
- ▶ Student copula available in dependence inference



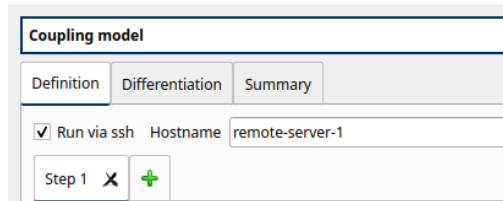
v19: Design of experiments – export & import

- ▶ The **underlying DOE** of any simulation analysis (central tendency, sensitivity, ...) can now be **exported as a Persalys DOE**
- ▶ Evaluations can be **imported from a CSV file** into an existing DOE
 - ▶ Allows running evaluations outside Persalys and re-importing results
- ▶ Monte Carlo is now the default choice in the DOE wizard



v19.1: Remote coupling via SSH

- ▶ Coupling models can now be executed on a **remote server via SSH**
- ▶ Requirements: passwordless SSH access to the target machine
- ▶ Persalys automatically:
 - ▶ Transfers input/output files
 - ▶ Runs the command on the remote server
 - ▶ Retrieves results
- ▶ Enables use of high-performance computing clusters without manual file transfers



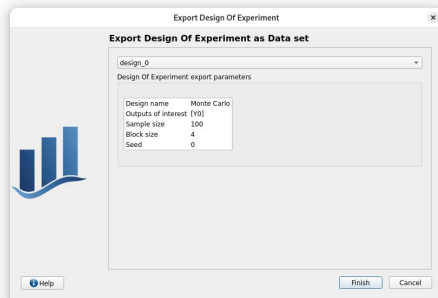
The screenshot shows a web-based configuration interface for a 'Coupling model'. At the top, there's a title bar 'Coupling model'. Below it are three tabs: 'Definition' (active), 'Differentiation', and 'Summary'. Under the 'Definition' tab, there is a checkbox labeled 'Run via ssh' which is checked. To its right is a label 'Hostname' followed by a text input field containing 'remote-server-1'. At the bottom left of the configuration area, there is a button labeled 'Step 1' with a red 'X' icon, and next to it is a green '+' icon.

v19 / v19.1: Regression improvements

- ▶ **Plain regression** analysis added (v19.1) – in addition to the existing stepwise linear regression
- ▶ Advanced parameters for **stepwise regression** (v19.1)
- ▶ Inference ergonomics (v19.1):
 - ▶ New button to copy the selected distribution to all other marginals at once
- ▶ **Histogram** distribution added (v19.0)
- ▶ Metamodel can be **exported as a Python model** (v19.1) – creates a physical model wrapping the metamodel

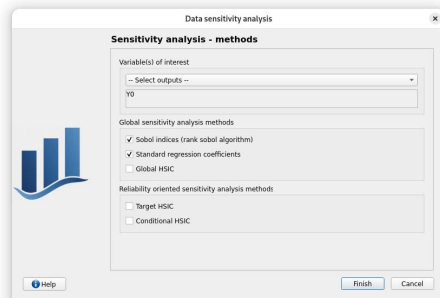
v20: Workflow reorganization

- ▶ Some analyses **moved from physical models to data sets**: SRC sensitivity and metamodel creation
- ▶ Recommended workflow:
 1. Create a design of experiment and evaluate it
 2. **Export it as a data set**
 3. Run all subsequent analyses (sensitivity, metamodel, ...) on the data set
- ▶ Simulation analyses can **export results** to a data model
- ▶ Monte Carlo analysis can now continue **even if some evaluations failed**



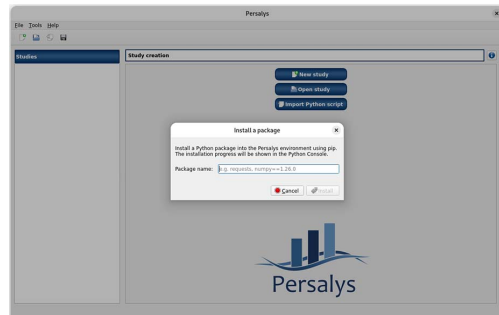
v20: HSIC sensitivity analysis

- ▶ New **HSIC (Hilbert-Schmidt Independence Criterion)** indices for sensitivity analysis
- ▶ Available on data sets
- ▶ Three modes:
 - ▶ **Global:** global sensitivity (screening)
 - ▶ **Target:** reliability-oriented, focus on tail events
 - ▶ **Conditional:** sensitivity conditioned on a critical domain
- ▶ Complements Sobol' indices, handles non-linear and non-monotonic models well



v20: Miscellaneous improvements

- ▶ **Install Python packages** button added in the interface
 - ▶ Wizard to install packages inside the Persalys Python environment without leaving the GUI
- ▶ **Data model type parameter** to help with results interpretation



Summary

Release	Highlights
18.0	Quantile analysis, data field model
18.1	Consistent MSE/ R^2 , CSV missing values
19.0	Data set sensitivity (Rank Sobol/SRC), DOE export/import
19.1	SSH remote coupling, regression, ergonomics
20.0	HSIC, workflow reorganization, package installer
20.1	Miscellaneous improvements, bug fixes

- ▶ Download: <https://persalys.fr>
- ▶ Forum: <https://persalys.discourse.group>

The end

Thank you!

Questions?